Land Use/Land Cover Transformation and Forecasting in Gorakhpur City Using GIS and Deep Learning Approaches

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Abstract

Rapid urbanization has significantly altered land use/land cover (LULC) patterns in medium-sized Indian cities. This study integrates Geographic Information System (GIS) techniques with deep learning models—Artificial Neural Network (ANN) and Multilayer Perceptron (MLP)—to analyze spatio-temporal LULC changes in Gorakhpur City using multi-temporal Landsat data for 2005, 2015, and 2025, and to predict future scenarios for 2035, 2045, and 2055. Supervised Maximum Likelihood Classification was used to generate LULC maps for five major classes. Change detection results indicate a 71.73% increase in built-up area and an 18.75% decline in agricultural land during 2005–2025. For prediction, ANN and MLP models were trained using transition probability matrices with a 70:30 training–validation split. The MLP model achieved higher accuracy (91.8%) and lower RMSE (0.096) compared to ANN (86.4%, RMSE = 0.142), demonstrating improved prediction performance. The proposed GIS–deep learning framework improves representation of nonlinear land transformation processes compared to conventional approaches. Future projections indicate continued urban expansion, agricultural land loss, and increasing encroachment into flood-prone and environmentally sensitive areas. The results provide a quantitative basis for sustainable urban planning and decision-making.
Keywords: *Land Use/Land Cover (LULC), Urban Expansion, Remote Sensing, Change Detection, Artificial Neural Network (ANN), Multilayer Perceptron (MLP), GIS Modeling, Urban Growth Prediction*

1. INTRODUCTION

Urbanization is a major driver of land use/land cover (LULC) transformation, particularly in developing countries. The global urban population has exceeded 55% and is projected to reach 68% by 2050 [1]. This expansion has resulted in large-scale conversion of agricultural land, forests, and water bodies into built-up areas, leading to environmental degradation, increased surface runoff, and urban heat island effects [2], [3].
Monitoring LULC dynamics is essential for understanding spatial transformation and supporting sustainable urban planning. Remote Sensing (RS) and Geographic Information Systems (GIS) enable consistent multi-temporal analysis of urban growth and land transformation patterns [4]–[6]. These techniques provide quantitative assessment of urban expansion and associated environmental impacts.
Recent advances in artificial intelligence have improved LULC prediction capabilities. Artificial Neural Networks (ANN) and Multilayer Perceptron (MLP) models can capture nonlinear relationships between historical land use patterns and future transitions [7], [8], and have demonstrated improved performance over conventional approaches such as Cellular Automata (CA)–Markov models [9]–[11].

However, existing studies are primarily focused on metropolitan regions and often rely on traditional modeling approaches or limited machine learning frameworks. In addition, comparative evaluation of ANN and MLP models for long-term LULC prediction in medium-sized cities remains insufficient, particularly in rapidly urbanizing regions such as eastern Uttar Pradesh.

This study addresses these limitations by integrating GIS-based LULC analysis with deep learning models for long-term urban growth prediction. The key contributions of this study are: (i) comparative evaluation of ANN and MLP models for LULC forecasting; (ii) integration of transition probability-based change detection with deep learning models; and (iii) application to a rapidly urbanizing medium-sized city (Gorakhpur) for multi-temporal prediction (2035–2055). The proposed framework provides a quantitative basis for sustainable urban planning, risk assessment, and evidence-based decision-making.

2. STUDY AREA

Gorakhpur City is located in eastern Uttar Pradesh, India (26°40′N–26°52′N and 83°15′E–83°33′E), within the Middle Gangetic Plain and the Rapti River basin, a tributary of the Ghaghara River system [12]. The region is characterized by flat alluvial terrain, fertile soils, and a humid subtropical monsoon climate, making it suitable for agriculture but highly vulnerable to seasonal flooding [13].

The population of Gorakhpur Urban Agglomeration increased from 0.62 million in 2001 to over 0.8 million in recent years, representing a growth of approximately 29%, indicating increasing urbanization pressure. This growth has contributed to significant expansion of built-up areas and conversion of agricultural land (see Table II).

The Rapti River flows along the western and southern boundaries of the city and plays a critical role in regional hydrology. Frequent flood events, combined with recent urban encroachment into low-lying and floodplain areas, have increased environmental vulnerability and flood risk [15].

Over the past two decades, Gorakhpur has experienced rapid peri-urban expansion and corridor-based development driven by transportation and infrastructure growth. These characteristics make Gorakhpur a suitable case study for analyzing spatio-temporal LULC dynamics and forecasting future urban growth using GIS and deep learning approaches.

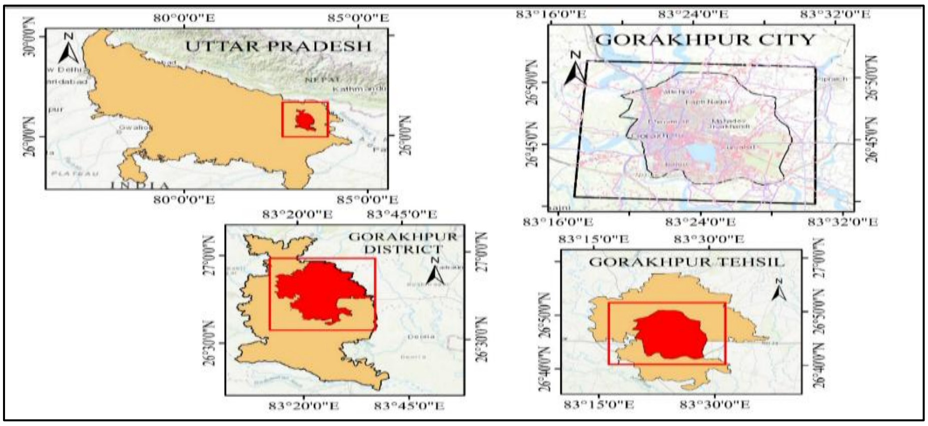

Fig 1. Location map of Gorakhpur City showing its spatial extent within Uttar Pradesh, Gorakhpur district, and tehsil boundaries.

3. DATA AND METHODOLOGY

3.1 Data Used

Multi-temporal Landsat datasets were used to analyze Land Use/Land Cover (LULC) changes in Gorakhpur City for the years 2005, 2015, and 2025. The details of the datasets are summarized in **Table I**. Landsat imagery was selected due to its temporal consistency, long-term availability, and suitability for regional-scale LULC analysis [17], [18]. All datasets have a spatial resolution of 30 m, which provides an appropriate balance between spatial detail and computational efficiency.
To ensure spectral consistency, images from the post-monsoon season (October–November) were used for all study years. The datasets were obtained from the USGS Earth Explorer platform and preprocessed using standard procedures, including radiometric, atmospheric, and geometric corrections, followed by clipping to the study area. The use of consistent Landsat datasets ensures temporal comparability and minimizes classification bias across different time periods.
All images were projected to the UTM coordinate system (Zone 44N, WGS 84 datum) to maintain spatial consistency across temporal datasets. These datasets form the basis for subsequent classification, change detection, and prediction modeling

Table I. Multi-temporal Landsat datasets used for LULC mapping in Gorakhpur City.

Year	Satellite	Spatial Resolution	Path/Row	Acquisition Period	Source
2005	Landsat 5 TM	30 m	142/41	Oct–Nov 2005	USGS Earth Explorer
2015	Landsat 8 OLI	30 m	142/41	Oct–Nov 2015	USGS Earth Explorer
2025	Landsat 8/9 OLI	30 m	142/41	Oct–Nov 2025	USGS Earth Explorer

3.2. Methodology Framework

The overall methodological framework adopted in this study is illustrated in Fig. 2. The approach integrates multi-temporal satellite data processing, LULC classification, change detection, and deep learning-based prediction.
Initially, Landsat imagery for 2005, 2015, and 2025 was preprocessed using standard radiometric, atmospheric, and geometric corrections. Supervised classification using the Maximum Likelihood algorithm was applied to generate LULC maps for each year. Classification accuracy was evaluated using confusion matrices, Overall Accuracy, and Kappa coefficient.
Post-classification comparison was performed to quantify LULC transitions between 2005 and 2025. A transition probability matrix derived from LULC changes was used to quantify class-wise conversion likelihood and served as input to the ANN and MLP models. The models were trained using a 70:30 training–validation split and optimized through backpropagation to minimize prediction error.

These models were used to capture temporal land transformation patterns and to predict future LULC scenarios for 2035, 2045, and 2055. Model performance was evaluated using accuracy metrics, including Overall Accuracy, Kappa coefficient, and RMSE.
This integrated GIS–deep learning framework enables quantitative assessment of past LULC dynamics and reliable prediction of future urban growth patterns.

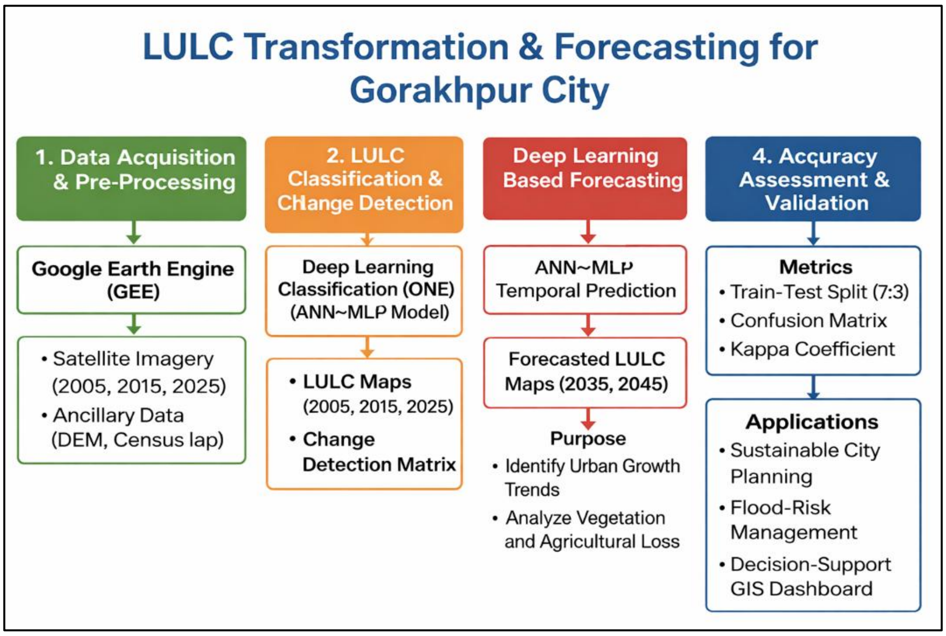

Fig 2. Methodological framework for LULC classification, change detection, and ANN–MLP-based prediction using multi-temporal Landsat data.

3.3 LULC Classification

LULC maps for 2005, 2015, and 2025 were generated using the supervised Maximum Likelihood Classification (MLC) algorithm, which accounts for class variance and co-variance. Five major classes were identified: agricultural land, built-up area, forest, wasteland, and water bodies. These classes were selected based on their dominance in the study area and their relevance to urban growth and environmental assessment.

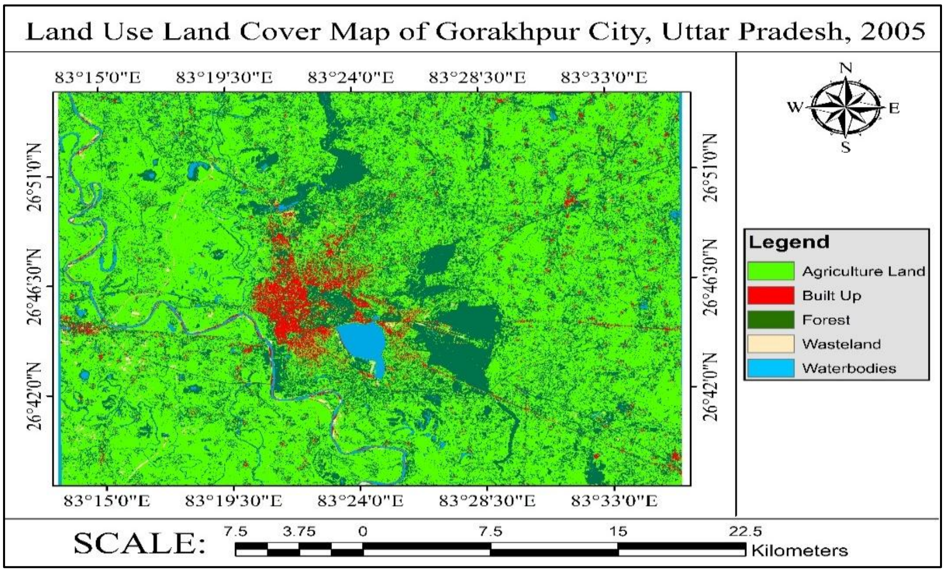

Fig 3. LULC map of Gorakhpur City for 2005 derived from Landsat 5 TM using MLC.

Training samples were selected through visual interpretation supported by high-resolution reference imagery. The classified LULC maps for 2005, 2015, and 2025 are shown in Figs. 3–5. The results indicate a progressive increase in built-up areas and a decline in agricultural land, with urban expansion concentrated around the city core and along major transportation corridors.

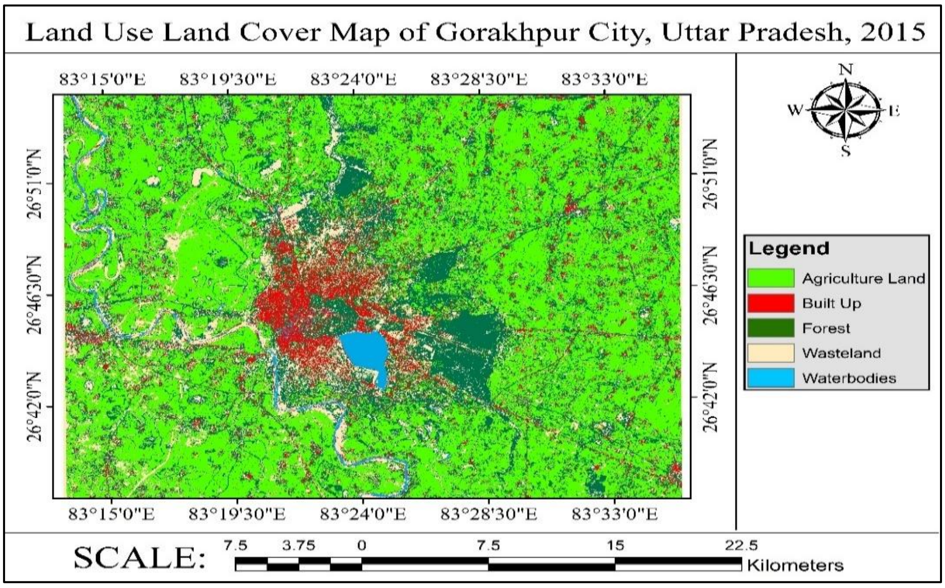

Fig 4. LULC map of Gorakhpur City for 2015 derived from Landsat 8 OLI using MLC.

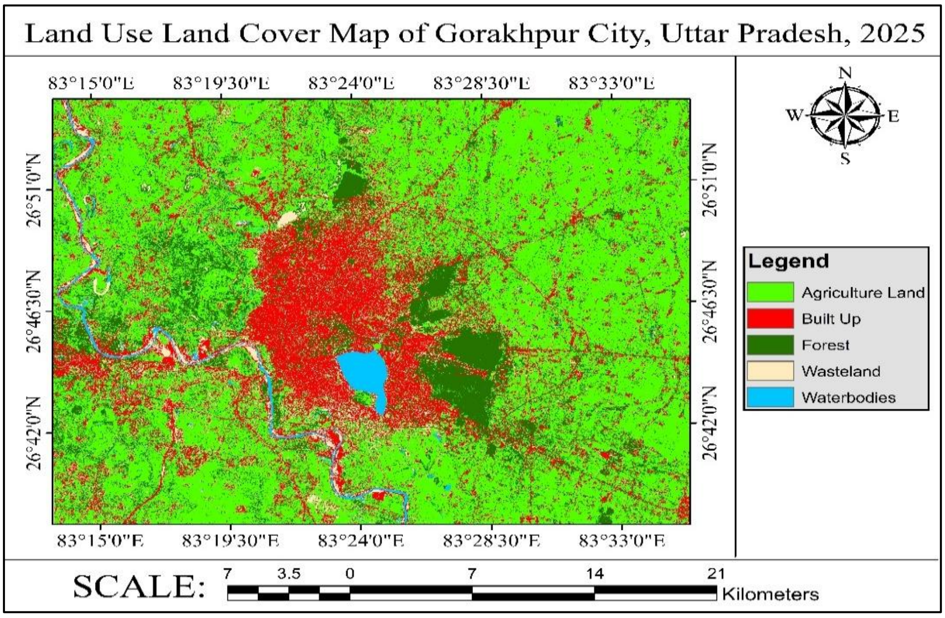

Fig 5. LULC map of Gorakhpur City for 2025 derived from Landsat 8/9 OLI using MLC.

Classification accuracy was evaluated using confusion matrices and standard metrics, including Overall Accuracy (OA), Producer's Accuracy (PA), User's Accuracy (UA), and Kappa coefficient [20]. The overall accuracy exceeded 85% for all years, confirming the reliability of the classification results. The detailed accuracy assessment is presented in Table II.

Table II. Accuracy assessment of LULC classification.

Year	Overall Accuracy (%)	Kappa Coefficient
2005	87.3	0.83
2015	89.6	0.86
2025	91.2	0.88

3.4 Change detection Analysis

Post-classification comparison was performed to quantify LULC changes between 2005 and 2025. The results indicate significant spatial transformation driven by urban expansion. Area statistics for each LULC class are summarized in Table III.

Built-up area increased from 42.8 km² to 73.5 km² (+71.73%), representing the dominant land transformation associated with rapid urban growth. In contrast, agricultural land declined from 145.6 km² to 118.3 km² (−18.75%), indicating conversion of peri-urban farmland into built-up areas. These changes indicate strong urbanization pressure, with built-up expansion occurring primarily at the expense of agricultural land.

Forest, wasteland, and water bodies exhibited relatively minor changes, suggesting localized variations rather than large-scale transformation. Wasteland showed moderate reduction due to its conversion into urban land use.

Overall, the results highlight a transition from an agriculture-dominated landscape to an urbanizing system, emphasizing the impact of sustained urban expansion on land resources.

Table III. LULC area statistics and change analysis for Gorakhpur City (2005–2025)

LULC Class	2005 Area (km²)	2015 Area (km²)	2025 Area (km²)	Net Change (2005–2025) (km²)	% Change
Agricultural Land	145.6	132.4	118.3	-27.3	-18.75%
Built-up Area	42.8	56.7	73.5	+30.7	+71.73%
Forest	18.4	19.2	17.9	-0.5	-2.71%
Wasteland	22.5	20.1	19.6	-2.9	-12.89%
Water Bodies	11.7	12.4	11.2	-0.5	-4.27%

3.5. ANN/MLP Prediction Model

Future LULC scenarios for 2035, 2045, and 2055 were predicted using Artificial Neural Network (ANN) and Multilayer Perceptron (MLP) models. Historical LULC maps for 2005, 2015, and 2025, along with transition probability matrices derived from change detection (2005–2025), were used as input variables.

Both models were designed to capture nonlinear relationships between past and future land use patterns. The ANN model consisted of an input layer, one hidden layer with 10 neurons, and an output layer. The MLP model employed a multi-layer feedforward architecture with two hidden layers (16 and 8 neurons) for improved feature representation.

The dataset was divided into training (70%) and validation (30%) subsets to ensure model generalization. Model training was performed using the backpropagation algorithm with adaptive optimization to minimize prediction error.

Model performance was evaluated using prediction accuracy, Root Mean Square Error (RMSE), and additional validation metrics such as coefficient of determination (R^2). The MLP model achieved higher accuracy and lower RMSE compared to ANN, indicating improved spatial consistency and reduced prediction noise. The results demonstrate that the MLP model provides more reliable long-term LULC predictions for the study area.

4. RESULT

4.1 LULC Transformation (2005–2025)

Spatio-temporal analysis of LULC from 2005 to 2025 indicates significant landscape transformation driven by rapid urbanization. Built-up area expansion is the dominant trend, characterized by urban densification in the city core and radial growth along major transportation corridors, reflecting infrastructure-driven development. These trends are quantitatively supported by the LULC statistics presented in Table III.

Agricultural land shows a consistent decline due to conversion of peri-urban farmlands into residential, commercial, and infrastructural uses, indicating increasing pressure on land resources. In contrast, forest and water bodies exhibit relatively minor changes, suggesting localized variations rather than large-scale transformation. However, encroachment into low-lying and floodplain areas indicates emerging environmental stress and potential flood risk.

Overall, the results demonstrate a transition from an agriculture-dominated landscape to an urbanizing system, driven by core densification, corridor-based expansion, and peri-urban conversion.

4.2 Future LULC Prediction for 2035

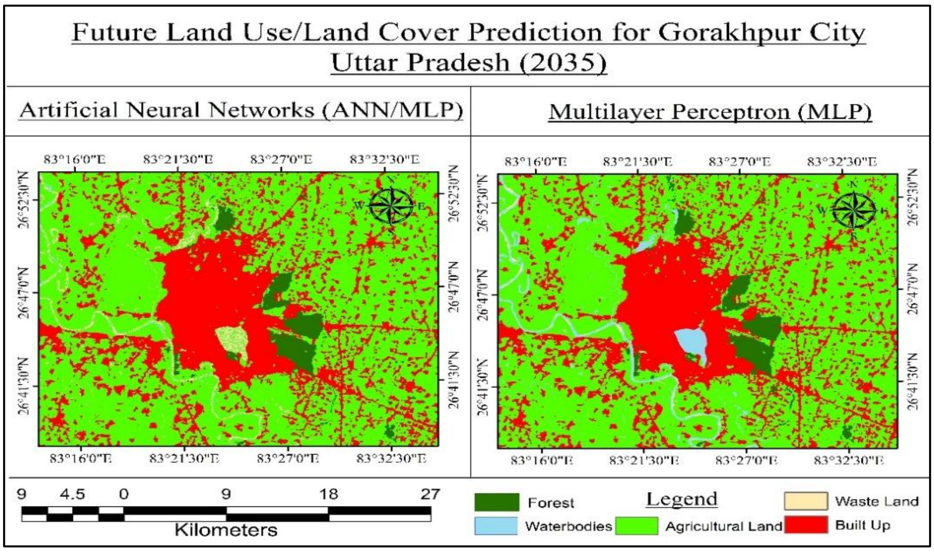

Fig 6. Predicted LULC map for 2035 using ANN and MLP models.

The predicted LULC scenario for 2035 is shown in Fig. 6. The results indicate continued expansion of built-up areas beyond the current urban boundary, particularly toward the southern and eastern regions dominated by agricultural land. The predicted increase in built-up area and corresponding decline in agricultural land are consistent with the observed trends in Table III, indicating continuation of historical urban growth patterns.

Urban growth exhibits a radial and corridor-based pattern along major transportation routes, leading to fragmentation and conversion of peri-urban agricultural land. The predicted expansion also indicates increasing encroachment into low-lying and water-adjacent areas, suggesting elevated flood risk and environmental vulnerability.

Comparison of model outputs shows that the MLP model produces smoother and more spatially consistent urban growth patterns, whereas the ANN model exhibits relatively fragmented predictions. Overall, the 2035 scenario reflects accelerated urban expansion, peri-urban transformation, and increasing pressure on environmentally sensitive zones.

4.3 Future LULC Prediction for 2045

The predicted LULC scenario for 2045 is shown in Fig. 7. The results indicate accelerated urbanization compared to 2035, with increased peripheral expansion and infill development within existing urban areas, reflecting urban consolidation. These trends are consistent with the LULC statistics presented in Table III.

Built-up areas exhibit more continuous and compact growth, particularly along established transportation corridors, indicating a transition from radial expansion to intensified urban structure. This expansion is accompanied by significant reduction of agricultural and open land, leading to fragmentation of peri-urban zones and decline in green spaces. This indicates a shift from expansion-driven growth toward consolidation-dominated urban development.

Model comparison shows that the MLP output produces smoother and more spatially coherent patterns, while the ANN model exhibits relatively fragmented urban patches. Overall, the 2045 scenario highlights intensified urban consolidation, increasing dominance of built-up areas, and growing pressure on land resources.

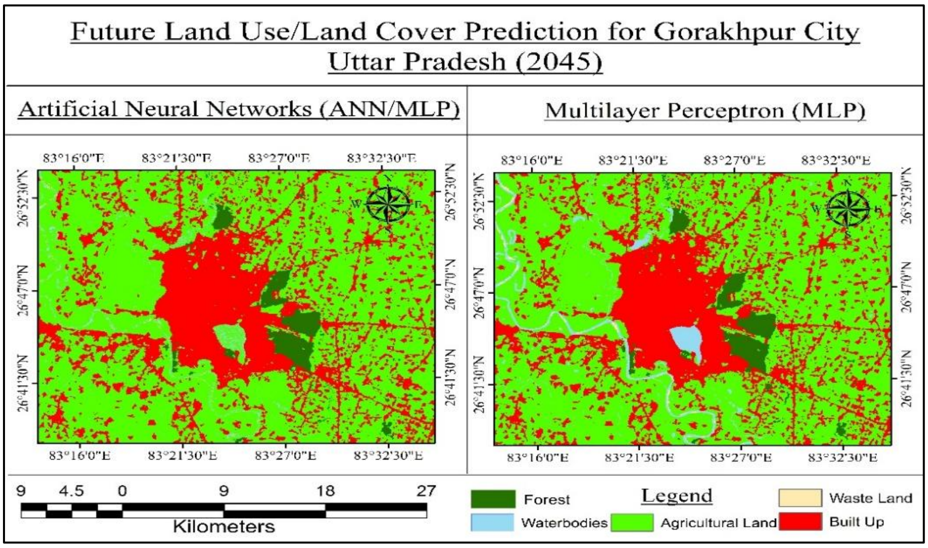

Fig 7. Predicted LULC map for 2045 using ANN and MLP models.

4.4 Future LULC Prediction for 2055

The long-term LULC scenario for 2055 is presented in Fig. 8. The results indicate dominant expansion of built-up areas across both urban and peri-urban regions, reflecting a transition toward a highly urbanized landscape. These trends are consistent with the LULC statistics presented in Table III. Agricultural land shows substantial reduction due to continued conversion into residential, commercial, and infrastructural uses, leading to fragmentation of peri-urban zones.

Urban growth patterns indicate increased consolidation, with previously dispersed development merging into continuous urban clusters. The predicted expansion into low-lying and environmentally sensitive areas suggests elevated flood risk and reduction in green spaces. This scenario represents a near-saturation stage of urban growth under current development trends.

Model comparison shows that the MLP output provides more spatially coherent and continuous urban patterns, whereas the ANN model exhibits relatively scattered growth.

Overall, the 2055 scenario highlights extensive urban dominance, land resource pressure, and the need for long-term planning interventions.
elevated flood risk and reduction in green spaces.

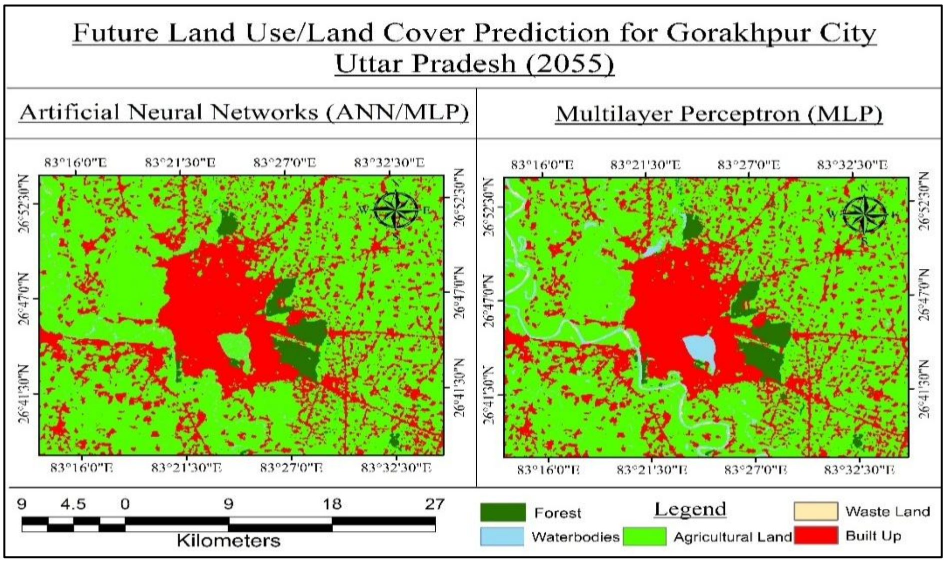

Fig 8. Predicted LULC map for 2055 using ANN and MLP models.

4.5 ANN vs MLP Performance Comparison

The predictive performance of ANN and MLP models was evaluated using prediction accuracy, Root Mean Square Error (RMSE), and spatial consistency metrics. The results are summarized in Table IV.
The MLP model achieved higher accuracy (91.8%) and lower RMSE (0.096) compared to the ANN model (86.4%, RMSE = 0.142), indicating improved prediction precision. In addition, MLP outputs exhibited more spatially coherent and continuous urban growth patterns, whereas ANN results showed relatively fragmented predictions. Model performance was further supported by additional validation metrics such as coefficient of determination (R^2).
The improved performance of the MLP model is attributed to its enhanced capability to capture nonlinear relationships and optimize feature representation during training. This highlights the superiority of the MLP model in capturing complex spatial patterns of urban growth. Overall, the results demonstrate that the MLP model provides more reliable and consistent predictions for long-term LULC forecasting.

Table IV. Performance comparison of ANN and MLP models for LULC prediction.

Model	Prediction Accuracy (%)	RMSE	Urban Spread Precision
ANN	86.4	0.142	Moderate
MLP	91.8	0.096	High

5. IMPLICATIONS FOR URBAN PLANNING

The observed increase in built-up area and decline in agricultural land indicate the need for controlled land-use zoning to regulate urban expansion. Zoning strategies should prioritize compact urban development and limit conversion of productive agricultural land in peri-urban areas.

The reduction of agricultural land highlights the importance of preserving agricultural buffer zones, which support food security and act as ecological transition areas. Protection of these zones can also help mitigate urban heat island effects.

Encroachment into low-lying and floodplain areas increases flood risk and urban waterlogging. Conservation of wetlands, ponds, and river corridors, particularly along the Rapti River, should be integrated into urban planning through strict land-use regulations in vulnerable zones.

Projected urban consolidation underscores the need for green belts and ecological corridors to maintain environmental quality, support biodiversity, and improve urban microclimate. These implications are directly derived from the observed LULC transformation and future prediction results presented in Section IV.

Implementation of these strategies is essential to mitigate environmental risks associated with uncontrolled urban expansion. Overall, the integration of GIS-based monitoring with deep learning-based prediction provides a quantitative framework for evidence-based urban planning and sustainable land management.

6. CONCLUSION

This study analyzed Land Use/Land Cover (LULC) changes in Gorakhpur City for 2005–2025 and predicted future scenarios for 2035, 2045, and 2055 using GIS and deep learning models. Results indicate a substantial increase in built-up area and a corresponding decline in agricultural land, reflecting rapid urban expansion and peri-urban transformation.

Comparative analysis shows that the Multilayer Perceptron (MLP) model achieved higher prediction accuracy and lower error than the Artificial Neural Network (ANN), with improved spatial consistency in simulated urban growth patterns. The predicted scenarios indicate continued urban expansion, agricultural land loss, and increasing encroachment into flood-prone and environmentally sensitive areas.

The key contribution of this study lies in the integration of transition probability-based change detection with ANN and MLP models for long-term LULC forecasting in a medium-sized city context. These trends highlight the need for controlled land-use planning, protection of agricultural buffers, and regulation of development in vulnerable zones.

Although the study provides reliable predictions, the accuracy of future scenarios depends on the quality of input data and model assumptions. Future research can incorporate higher-resolution datasets and advanced deep learning architectures to further improve prediction performance.

Overall, the proposed GIS–deep learning framework provides a quantitative basis for long-term urban growth assessment and supports evidence-based decision-making for sustainable land management in rapidly urbanizing cities.

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